

— HURRICANE AERIAL DETECTION AND EMERGENCY SEARCH



HADES

Autonomous AI drones that find hurricane survivors in real time, cutting search time from hours to minutes.

Andy · Soham · Aarush

Durham, North Carolina

— WHAT IS IT

An autonomous drone system that uses AI to locate hurricane survivors in real time, and puts them on a **live map** for rescue coordinators.

Autonomous

Flies wide-area sweeps. No spotter required.

AI-Powered

Detects people and computes their real-world coordinates.

Under \$700

Versus \$10,000+ per helicopter flight hour.

— WHY THIS MATTERS

Nearly half of all hurricane deaths happen **after** the storm, during the rescue window, when help is too slow and too scarce.

- ◆ Helicopters cost **\$10,000+ per flight hour**, cover one zone at a time, and rely on a spotter whose focus lapses after about 20 minutes.
- ◆ During Katrina, **40+ helicopters flew nonstop** and still could not cover everything.
- ◆ **Hurricane Helene, Sept 2024:** 108 deaths statewide. Towns in western NC went silent for **72+ hours**.

FLOOD / DISASTER PHOTO
DROP IN ONE POWERFUL IMAGE

We did not build this in a vacuum. It started with people who **lived it**.

- ◆ **Ryan Cannon and Arya Patel**, friends from our dorm hall, both from western NC. Their firsthand accounts of Helene shaped every core design decision.
- ◆ Studied **Coast Guard data from Katrina** and **FEMA post-storm assessments** to learn where aerial rescue breaks down at scale.
- ◆ Pressure-tested our model with **Dr. Sakaguchi** [ML PhD, Duke; former DoD contractor at CoVar Technologies].
- ◆ Worked with **Taylor Gibson** [Exec. Director, Ryden Program for Innovation and Leadership in AI] on deployment realities.
- ◆ Began working with **NC emergency management officials**, whose feedback is actively shaping our coordinator app and detection thresholds.

Three steps: **Deploy, Detect, Coordinate.**

01 / DEPLOY

Fly the sweep

A drone flies a wide-area sweep, streaming live video to a ground-station laptop over a digital video link.

02 / DETECT

Find the people

A **YOLO11s** model flags people on rooftops, near submerged vehicles, and in open water. **Temporal aggregation** across frames keeps real survivors and filters one-frame false positives.

03 / COORDINATE

Place them on a map

Our pipeline turns each detection into GPS coordinates using onboard sensors only. **No LiDAR. No RTK.** Locations stream to a live survivor map.

The stakes are literally life and death, so ethics is a **design requirement**, not an afterthought.

- ◆ **Sensitivity over silence.** A missed survivor is worse than a false alarm, so we tune to flag more. The AI surfaces information; a human coordinator always makes the final call.
- ◆ **Auditing for bias.** Disaster datasets underrepresent some skin tones, clothing, and lighting. We are auditing and supplementing our training data, guided by Dr. Sakaguchi.
- ◆ **No false certainty.** Every detection ships with two separate confidence scores and an honest uncertainty radius. We never paint a false-precision pin.
- ◆ **Scoped deployment.** Designed to operate only in officially designated disaster zones, alongside authorized agencies, never as a civilian surveillance tool.

Three high schoolers, each leading the part they **actually know**.

PHOTO

DESIGN

Andy

Years of FTC robotics. CADed the carbon-fiber airframe from scratch in Onshape, integrating every component before a single cut.

PHOTO

MANUFACTURING

Soham

UAV and drone competition experience. Owns motor configuration, propeller selection, and flight behavior.

PHOTO

AI & PROGRAMMING

Aarush

Four years of FRC competitive programming. Built the detection pipeline, the localization algorithm, and the coordinator app.

What makes us right for this: our connection to the problem is **personal**. Ryan and Arya's accounts of Helene drove everything we built.

We do not just claim it works. We **measured** it.

0.55

Detection recall, YOLO11s @ 960px
held-out HERIDAL; 0.63–0.69 after
confirmation

~10 m

Nominal localization error
1.5–5 m near-nadir; reported with
uncertainty

63 fps

Detection throughput, on a laptop, no GPU
22 ms in-app latency; budget is 120 ms

- ◆ **Recognition.** 1st place, NC Science Fair [technology category]. Collaborated with MIT CSAIL and Duke; supported by the Bowman Brockman Fund, Polymaker, and DolphinRC.
- ◆ **Community.** NC emergency management feedback is actively shaping the coordinator app and detection thresholds.

— WHAT'S NEXT

The single greatest determinant of survival after a disaster is the time between being stranded and being found.
HADES attacks that directly.

NEXT

Move from recorded-data validation to live flight integration.

NEXT

Deepen the NC emergency management partnership toward a field pilot.

Half of hurricane deaths come after the storm. A **\$700 drone** that finds people in minutes, not days, changes who makes it.



HADES

Thank you

Andy · Soham · Aarush